AUTUMN **MID SEMESTER EXAMINATION-2025**

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Design and Analysis of Algorithms
[CS30001]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give theiranswers in their own words as far as practicable and all parts of a question should beanswered at one place only.

1. Answer all the questions. [1 X 5]

   a) Explain Big-O notation with its formal definition. Why is it important in algorithm analysis?

   b) Consider the following function:

```
int compute (int n)
{     int i=2;
      while (i<n)
            i=i^2;    }
```

   Which of the following represents the time complexity of the algorithm?

   (i) O (n)
   (ii) O ($\log_2$ n)
   (iii) O (n^2)
   (iv) O ($\log_2 \log_2$ n)

   c) An unsorted list contains n distinct elements. The number of comparisons to find an element in this list that is neither maximum or minimum is:

   (i) O(n $\log_2$ n)
   (ii) O(n)
   (iii) O($\log_2$ n)
   (iv) O(1)

   d) Compare the asymptotic growth of $n^{logn}$and $(logn)^n$. Which one grows faster, and why? Rewrite each function in terms of exponentials of base 2 to justify your answer.

   e) Consider a max-heap with n elements implemented as an array. The maximum element is at index 1. What is the minimum and maximum possible number of elements in the last level of the heap?

2. Write the recurrence relation T(n) and solve it using the recursion-tree method for the quicksort algorithm when the pivot chosen is the (n/4)th smallest element of the array. [5]

3. Obtain optimal Huffman codes for a set of characters S : <a,b,c,d,e,f> with the following frequencies F:<6, 4, 12, 15, 23, 40>. Find the the % of save in the file size using Huffman code with respect to equal length encoding representation when the file size is 2000 characters. [5]

4. Following the recent 50% tariff imposed by Mr. Trump on goods exported from India to the USA, Mr. Modi is compelled to visit the capitals of other countries in search of new export opportunities. The air distances between these capitals are represented by an undirected graph, where each vertex denotes a capital and the cost of each edge corresponds to the direct distance between two capitals. Using Dijkstra's shortest path algorithm, a predecessor array pred[ ] is computed. In this array, the index corresponds to a vertex, and the value indicates the previous vertex in the shortest path from the source (New Delhi) to a destination capital. During the journey, Mr. Modi loses weight at every takeoff: For the first takeoff, he loses 1 kg. For each subsequent takeoff, the weight loss doubles the previous loss. Write an algorithm to compute the total weight lost (in kg) by Mr. Modi during his shortest-path journey from New Delhi (source) to another country's capital (destination). Assume that in the path, the flight lands at all intermediate capitals lying between the source and the destination. Also, analyze and state the best-case and worst-case execution time of your algorithm for n countries (vertices).
[5]

5. a) Solve the recurrence relation

$T(n) = 3T\left(\frac{n}{3}\right) + n^2$, if n>1 and T(3)=1 [2.5]

b) Using the job scheduling deadlines find the optimal schedule for the following jobs, a, b, c, d, e, f, g, h, i. where Profits:<15, 20, 30, 18, 18, 10, 23, 16, 25>, and deadlines:<7, 2, 5, 3, 4, 5, 2, 7, 3>. [2.5]

*** Best of Luck ***